Technical pack
The Operator 35 L

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A matte black 72-hour emergency backpack for adults, sold worldwide. The factory makes the empty bag with its internal pouch set, sleeves and labels; contents are packed by us in the UK. **The brief: the most premium look and feel possible on a value material.** Premium comes from finish, not fabric cost: everything matte black, clean edges, structured shape, branded rubber details (finish checklist, page 2).

SHEET	CONTENTS	WHAT IT COVERS
1	**Exterior views**	520 H × 320 W × 220 D mm, 35 L. Name window by the handle, left bottle pocket, Morse SOS laser-cut slots on the right panel, hip belt, whistle sternum buckle, grab handle, compression straps, black reflective piping
2	**Interior and pouch set**	Lid mesh pocket, elastic loop row, card sleeve, A5 document sleeve, four colour-coded pouches with factory-printed names, sewn-through pouch rails with press-stud straps
3	**Docking system**	Four hidden female buckles on the front so our future child, baby and pet bags clip on. Must be built into every Operator now
4	**Branding, hardware, materials**	Rubber patch 70 × 40 mm, Morse cut, blue tag, rubber zip pulls, name card, inside label, colours, full material specification
5	**Bill of materials, QC, packing**	Quantities per bag, construction standards, pull, drop and water tests, packing, terms

What we need from you

1. **Unit price for 300 Operators**, FOB and DDP London, in the specification on Sheet 4.
2. Option prices, quoted separately: (a) YKK AquaGuard on the main zip only; (b) an individual matte black printed mailer box per bag.
3. One pre-production sample: cost, lead time, and whether the fee is refunded against the order.
4. Production lead time for 300 pieces after sample approval.
5. One-off tooling: rubber patch mould, rubber zip-pull mould, laser-cut Morse pattern.
6. Shell fabric swatches: 2 or 3 matte black options for us to choose from, with test reports and REACH compliance.

Contact

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Premium on a value material: the finish checklist

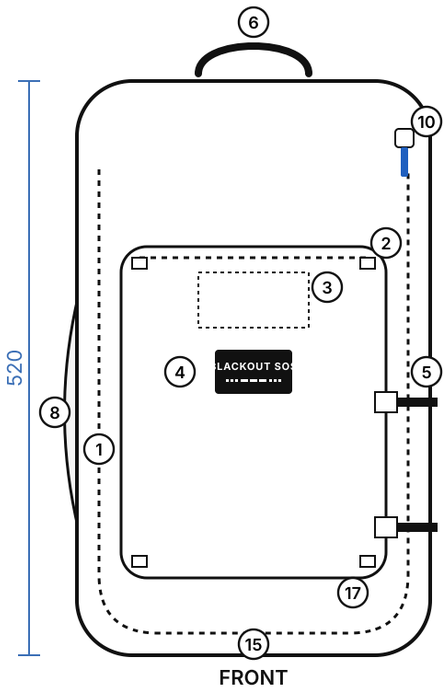
Every point below is a pass/fail item at sample approval and at pre-shipment inspection.

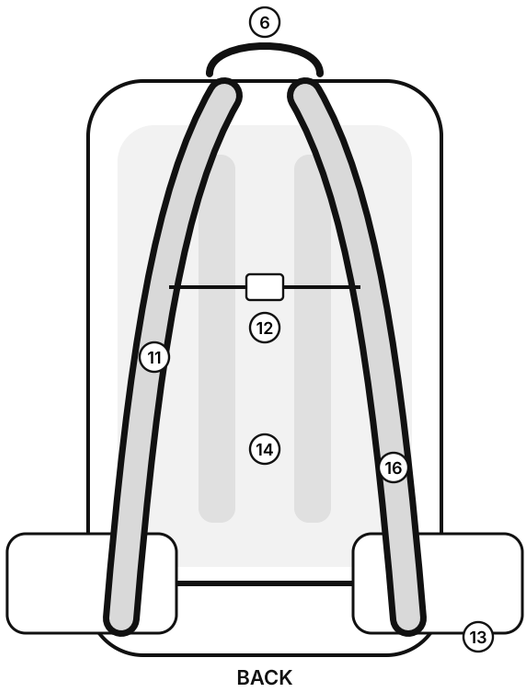
1. **One true black.** Shell, base, webbing, thread, zip tape, sliders, buckles and lining all match under daylight: no brown, blue or grey cast between parts. The only colour on the outside is the blue tag and side loop.
2. **Matte, never shiny.** 900D solution-dyed polyester with a fine matte twill face; the coating goes on the inside. No glossy PU face, no visible sheen.
3. **Holds its shape.** 10 mm EVA back panel and a 2 mm PE stiffener in the front panel so the bag stands upright on its own and the front stays flat with clean lines, empty or full.
4. **Branded rubber details.** Debossed black rubber zip pulls on every slider, the rubber front patch, and the laser-cut Morse SOS panel: these carry the premium look.
5. **Matte black metal sliders** on every external zip; matte black acetal buckles; nothing silver or glossy.
6. **Clean edges.** All internal seams bound with 20 mm tape, webbing ends heat-cut and folded under, no loose threads, no exposed foam, straight even stitching at 8–9 stitches per inch, double-needle on straps.
7. **Black reflective piping** on strap edges and the handle: invisible by day, bright in a torch beam.
8. **Hidden docking buckles** recessed under the front-pocket piping, so the front reads as one clean panel.

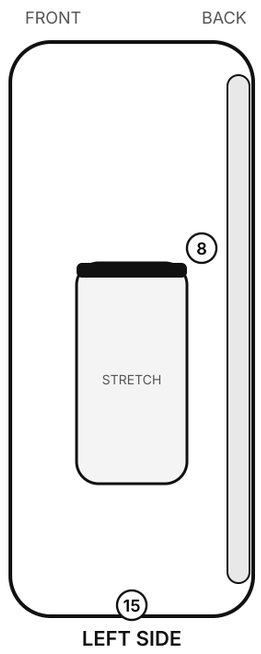
Concept render for look, finish and proportion only. All dimensions, materials and construction are as drawn on the sheets that follow. No camouflage, no visible MOLLE, no bright branding.

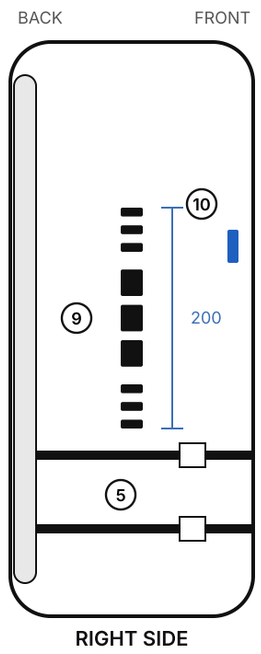
Sheet 1 — Operator 35 L · exterior views

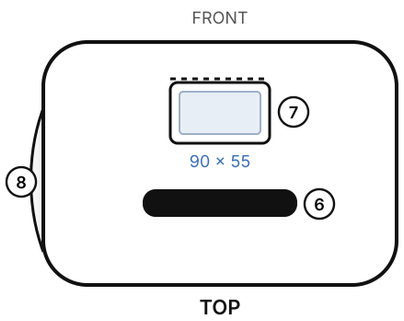
Scale 0.6 px per mm on screen · all dimensions in mm · colour: black throughout, tag blue











Overall
H 520 × W 320 × D 220 mm · 35 L nominal, 31 L usable
Empty weight target 1.3 kg · load rating 10 kg
Carry-on size: fits 55 × 40 × 23 cm

Build
Shell 900D polyester, matte twill, TPU coated, jet black
Base 1680D ballistic polyester · lining 210D ripstop
Main zip #8 water-resistant coated coil, 2 sliders, storm flap
Pocket zips #5 coated coil · matte black acetal buckles
Seams: taped on main body · bar-tacks at every strap anchor

1 Clamshell main opening, #8 water-resistant coated coil zip, two matte black metal sliders, storm flap

2 Front organiser pocket 240 × 300 mm, #5 coated coil zip, mesh inner

3 Loop-Velcro field 100 × 50 mm, black, for the customer's own patch

4 BLACKOUT SOS rubber patch 70 × 40 mm, black-on-black, sewn (Sheet 4)

5 Compression straps, right side only: 20 mm webbing, 20 mm side-release buckles, at 290 and 400 mm from top

6 Grab handle: 25 mm webbing, 140 mm, padded sleeve, bar-tacked both ends

7 Name-tag window 90 × 55 mm on top panel beside the handle: #5 zip, clear TPU window, holds an 85 × 54 mm card

8 Side bottle pocket, left: 100 × 200 mm heavy stretch mesh, elastic cuff, drain eyelet, fits 750 ml to 1 L

9 Morse SOS laser-cut slots, right side panel, 200 mm tall, black mesh backing (Sheet 4)

10 Colour tag 60 × 15 mm looped on the main zip pull, plus 10 × 30 mm loop on the right side; blue (Sheet 4)

11 Shoulder straps 60 mm, S-curve, 10 mm EVA foam, mesh face, 40 mm ladder-lock adjusters

12 Sternum strap 20 mm on rail, buckle with built-in whistle

13 Hip belt: 40 mm webbing, padded wings 150 × 90 mm, stows behind the back panel

14 Back panel: 10 mm EVA, two airflow channels, mesh face

15 Base panel 1680D ballistic polyester, PU coated, no feet

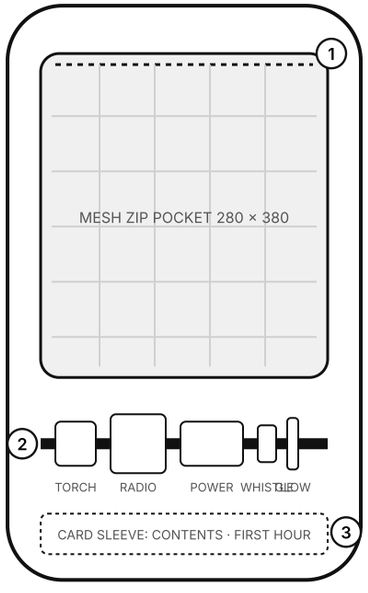
16 Black reflective piping 3 mm along strap edges and handle

17 Docking points: four 20 mm female side-release buckles on 25 mm webbing tabs at the front-pocket corners, centres 220 mm across and 280 mm down, recessed under the pocket piping so they don't show. Takes one of our future child, baby or pet bags at a time, 10 kg max (Sheet 3)

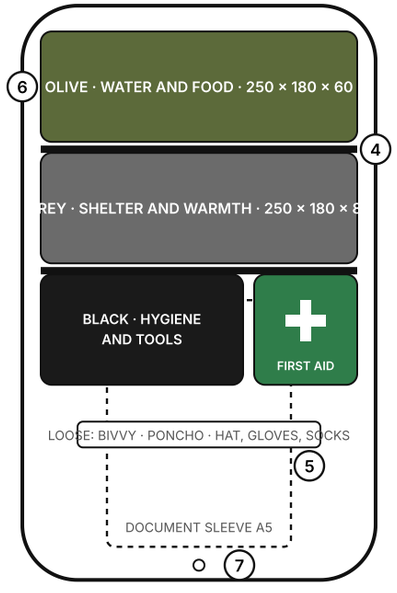
Not shown: hydration port omitted; no external MOLLE; all hardware black; 2 mm PE stiffener inside the front panel; all thread black.

Sheet 2 — Operator interior, packing map and pouch set

Lining black 210D polyester ripstop



LID (inside of front panel)



MAIN BODY (back wall)

Interior callouts

1 Lid mesh pocket 280 × 380 mm, black polyester mesh, #5 coil zip across the top, holds the lid items and the cards

2 Elastic loop row: 25 mm black elastic, five loops sized torch · radio · power bank · whistle · glow sticks, bar-tacked between loops

3 Card sleeve 150 × 60 mm, clear TPU, under the loop row: contents card and first-hour guide

4 Pouch rails: two rows of 25 mm nylon webbing across the back wall, bar-tacked every 50 mm to make loops. Each rail is sewn through the shell back panel with a 40 × 20 mm reinforcing patch behind every bar-tack, and its ends are caught in the side seams; never sewn to the lining alone. Each pouch has two 25 mm webbing straps on its back: the strap threads behind a rail loop and closes on itself with a black metal press stud (15 mm, ring-spring). Pouches lift out in one pull for grab-and-go and go back in seconds. Hook-and-loop (Velcro) is not used for attachment

5 Document sleeve A5 (148 × 210 mm), TPU-lined, Velcro flap, sewn flat to the back wall behind the pouches

6 Pouch packing map, top to bottom: olive, grey, then black beside green. Loose soft items sit on top of the pouches

7 Drain eyelet, 8 mm brass, black, in the base

8 Main compartment lining 210D polyester ripstop, black; lid lining the same with the mesh pocket over it

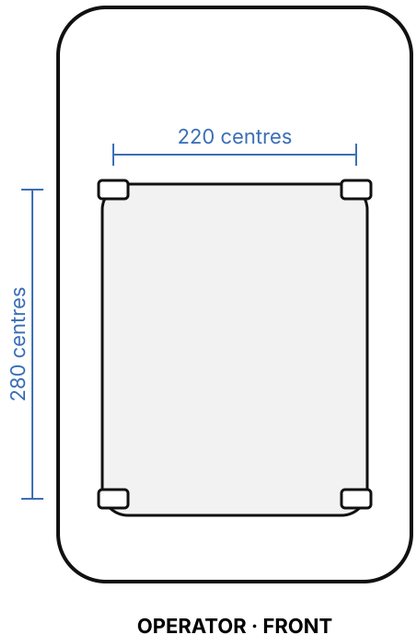
9 Two 20 mm webbing loops on the back wall at the top for the hydration bladder hook (no port, loops only)

Pouch	Colour	Size mm (W × H × D)	Holds	Construction
Water and food	Olive 210D ripstop	250 × 180 × 60	Water pouches, rations, comfort pack, tablets, spork	Boxed pouch, #5 coil zip on three sides, 20 mm grab tab, pouch name screen-printed by the factory on the front in white, 8 mm capitals (WATER AND FOOD · SHELTER AND WARMTH · HYGIENE AND TOOLS · FIRST AID), plus a 60 × 20 mm clear label window for our contents card, two 25 mm press-stud straps on the back for the rails (item 4), lining 210D black
Shelter and warmth	Grey 210D ripstop	250 × 180 × 80	Foil blankets, hand warmers; bivvy, poncho, hat, gloves and socks sit loose on top	
Hygiene and tools	Black 210D ripstop	250 × 180 × 60	Wipes, sanitiser, toilet roll, toothbrush, pads, bags, multi-tool, tape, cord, ties, ferro rod, pen	
First aid	Green 210D with white cross	180 × 120 × 60	First aid kit contents, trauma dressing, medication pouch, masks, gloves	
Lid mesh pocket	Black mesh	280 × 380 flat	Head torch, radio, power bank, cable, whistle, glow sticks, batteries	Sewn into the lid lining, #5 coil zip
Document sleeve	Black TPU-lined	A5, 160 × 220	Document pouch, map, contacts card, cash	Sewn to the back wall, Velcro flap

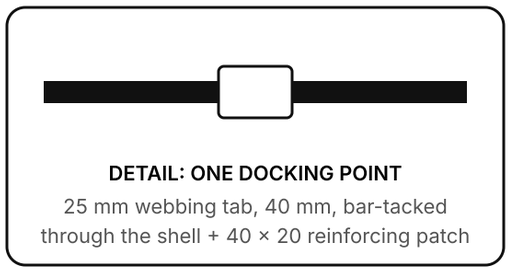
Sheet 3 — Docking system (piggyback points)

Scale 0.6 px per mm · dimensions in mm

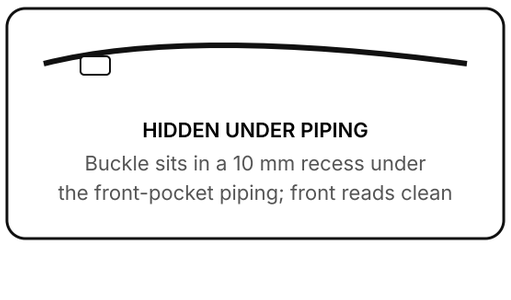
Every Operator carries four hidden female buckles on its front. Our smaller bags, coming later (child, baby and pet), have four matching male buckles on their backs and clip straight on, so one adult carries two bags hands-free. **These buckles must be built into all 300 Operators now**; the smaller bags will be made to this exact spacing.



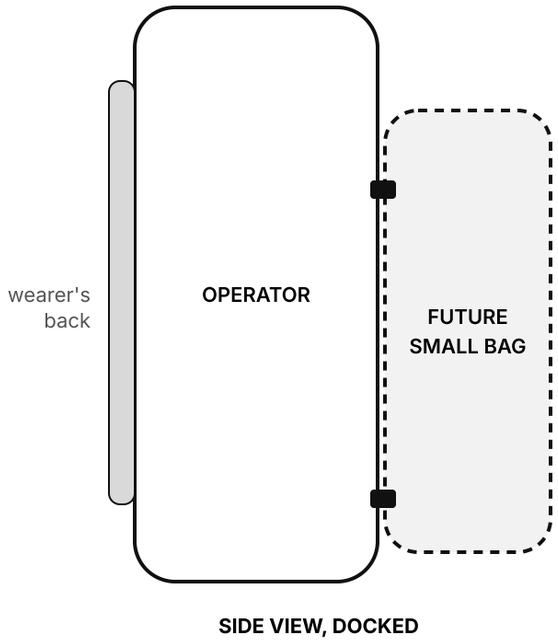
OPERATOR · FRONT
4 × 20 mm FEMALE side-release buckles
on 25 mm tabs at the front-pocket corners



DETAIL: ONE DOCKING POINT
25 mm webbing tab, 40 mm, bar-tacked through the shell + 40 × 20 reinforcing patch



HIDDEN UNDER PIPING
Buckle sits in a 10 mm recess under the front-pocket piping; front reads clean



SIDE VIEW, DOCKED
Small bag clips to the Operator's outer face
(dashed: not part of this order)

Rules

One docked bag at a time
Max docked load 10 kg
Spacing ±2 mm on every bag: future bags must fit all Operators made now

Tests on the Operator

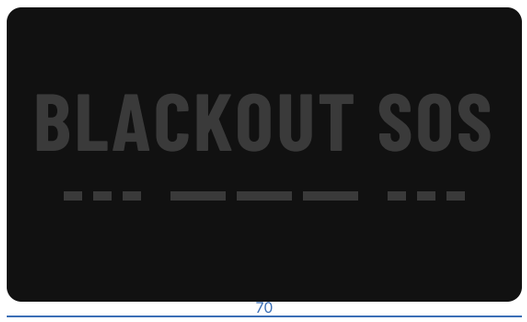
Each docking tab: 25 kg pull, no stitch failure
Buckle: 500 clip/unclip cycles
Every bag checked on a 4-buckle jig (below)

Please make a simple test jig (a flat board with four 20 mm male buckles at 220 × 280 mm centres) and use it to check docking on every Operator at inspection; send one jig with the sample.

Sheet 4 — Branding, hardware and materials

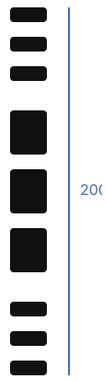
Artwork drawn at 4 px per mm · wordmark font Barlow Condensed Bold, all capitals · factory to supply strike-offs of every item

A · Front patch 70 × 40 mm



Matte black PVC rubber patch, 2 mm base, letters and Morse raised 1 mm in the same black; the mark reads only by its sheen. Cap height 9 mm. Morse: dots 2.5 × 1.2 mm, dashes 7.5 × 1.2 mm, 1.5 mm between marks, 4 mm between groups, 4 mm below the letters. Corner radius 2 mm. Sewn on with black thread 2 mm from the edge.

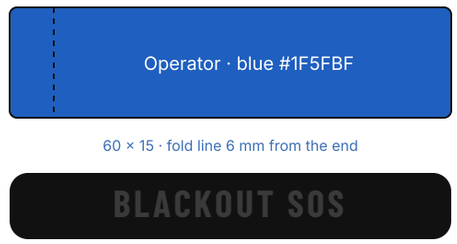
B · Morse SOS laser cut, side panel



Bag	Slot W	Dot	Dash	Gap / group	Total
Operator	20	8	24	8 / 16	200

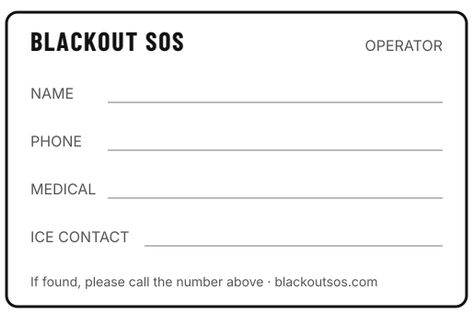
All in mm. Cut through the coated shell only, 2 mm corner radius, edges heat-sealed. Backed inside with black polyester mesh bonded 10 mm beyond the pattern on all sides, then the lining. Centred on the panel width; top of pattern 150 mm from the top seam.

C · Colour tag and zip pull



Tag: woven polyester, high density, 60 × 15 mm, colour both faces, folded and lark's-headed through the slider so the customer can swap it. Side loop: 10 mm webbing in the same colour, 30 mm loop, bar-tacked. Zip pull: black TPR rubber 60 × 12 × 3 mm, wordmark debossed 0.5 mm, 4 mm cap height, on every slider.

D · Name card 85 × 54 mm (printed, supplied by us)



350 gsm card, matt laminated, write-on with a ballpoint. Ships in the window; two spares in the lid sleeve.

E · Inside label 60 × 40 mm (woven, sewn into the back wall seam)



Black ground, white and grey weave. Label reads OPERATOR 35 L · MAX LOAD 10 kg. No QR code. A separate care and composition label to UK textile rules sits beside it.

F · Colours

Shell, webbing, hardware, thread — black. Match the shell swatch we approve; no blue or brown cast.

Blue tag, Operator — #1F5FBF, close to Pantone 2935 C

Olive pouch — #5C6A3A · **grey pouch** #6B6B6B · **green first-aid pouch** #2F7D4A with white cross

Pantone references are the nearest match for the factory's lab-dip; we approve on physical swatches, not screen. Tags must pass 20 washes without bleeding onto the black shell.

Material or hardware	Specification	Where	Accepted alternative
Shell fabric	900D polyester, high-tenacity, solution-dyed jet black, fine matte twill face (no shine), TPU coated 0.08 mm on the back, water column 3,000 mm or better	All outer panels	Equivalent "Cordura-look" matte polyester; send swatches
Base fabric	1680D ballistic polyester, PU coated, black	Base panel	1000D polyester, PU coated
Lining	210D polyester ripstop, PU coated, black; pouches dyed olive, grey, green, black	Compartments, pouches	None
Stretch fabric	Heavy stretch Lycra-type mesh, black	Bottle pocket	None
Mesh	Polyester 3D spacer mesh 4 mm (back panel, strap faces); flat polyester mesh (lid pocket, Morse backing), black	Back, straps, lid	None
Foam	EVA closed-cell, 10 mm back and straps, 5 mm hip belt and grip	Padding	PE foam of the same density
Zips	Water-resistant coated (PU) coil zips with storm flaps: #8 main, #5 pockets; black tape, matte black metal sliders	All external zips	Quote YKK AquaGuard on the main zip only as an option
Zip pulls	Black TPR rubber 60 × 12 × 3 mm, wordmark debossed, on every slider	Every zip	None: this is a key premium detail
Buckles and adjusters	Acetal, matte black, quality grade (ITW Nexus/Duraflex-equivalent): 20 mm side-release (whistle type on sternum, 4 female docking), 25 mm ladder locks, 40 mm hip belt	As Sheet 1	Generic acetal passing the same load tests
Webbing	Nylon, black, tight weave: 20 mm (straps, docking tabs), 25 mm (handle, rails), 40 mm (hip belt); elastic 25 mm black; all ends heat-cut and folded	All	Polypropylene not accepted
Reflective	Black reflective piping 3 mm (black by day, silver in a beam)	Strap edges, handle	None
Windows	Clear TPU 0.5 mm, cold-crack resistant to −20 °C	Name window, card sleeve, pouch label windows	Clear PVC 0.5 mm
Thread	Bonded polyester or nylon #69 (Tex 70), black, UV stable; #138 on handle	All	None
Seam sealing	Seam tape on all shell-to-shell seams of the main compartment	Main body	Welded seams if the factory offers them

Sheet 5 — Bill of materials, construction, QC and packing

Quantities per bag · factory to confirm consumption from its own patterns

COMPONENT	PER OPERATOR
Shell: 900D polyester, matte twill, TPU coated, jet black	1.2 m²
Base: 1680D ballistic polyester, PU coated	0.1 m²
Lining: 210D polyester ripstop, black · pouch fabric dyed olive, grey, black, green	1.4 m² + 0.6 m²
Stretch mesh (bottle pocket)	1 pocket
3D spacer mesh (back, straps) + flat mesh (lid pocket, Morse backing)	0.25 + 0.2 m²
EVA foam 10 mm (back panel, 2 straps) · 5 mm (hip belt, grip)	as listed
PE stiffener 2 mm, front panel, sewn into a sleeve	1
Main zip #8 water-resistant coated coil, 2 matte black metal sliders, storm flap	1,100 mm
Pocket zips #5 coated coil (front pocket, name window)	2
Internal zips #5 coil (lid mesh pocket; 4 pouches, three sides each)	1 + 4
Rubber zip pulls, debossed wordmark	9 (5 external, 4 pouches)
20 mm side-release buckles, matte black acetal	2 compression + 4 female docking
20 mm sternum buckle with built-in whistle	1
25 mm ladder locks · 40 mm hip-belt buckle	2 · 1
Webbing, black nylon: 20 mm · 25 mm · 40 mm	1.5 m · 0.9 m (incl. rails, docking tabs) · 1.2 m
Elastic 25 mm · black reflective piping 3 mm	0.4 m · 2.5 m
Press studs 15 mm, matte black metal, ring-spring (pouch straps)	8
Clear TPU 0.5 mm: name window 90 × 55, lid card sleeve, pouch label windows	1 + 1 + 4
Pouches: olive, grey, black, green with white cross; names screen-printed in white	4
Document sleeve A5, TPU-lined, Velcro flap	1
Rubber patch 70 × 40 · loop field 100 × 50 · blue tag 60 × 15 · blue side loop	1 each
Labels: inside woven · care and composition · drain eyelet 8 mm	1 · 1 · 1
Name card 85 × 54 mm	supplied by us

Construction

Lockstitch 8–9 stitches per inch; double-needle on straps; #138 thread on the handle. 42-stitch bar-tacks at strap tops and bottoms, handle ends, hip-belt anchors, compression straps, pouch-rail loops and docking tabs, each through the shell with a reinforcing patch. Main-compartment seams taped; all internal raw edges bound with 20 mm tape. Storm flap 20 mm over every external zip. Laser cut after coating, edges heat-sealed, mesh bonded behind. Webbing ends heat-cut and folded under.

Tests (sample and pre-shipment)

Water: closed bag under a shower for 10 minutes, no water inside the main compartment or name window. Load: handle 30 kg static for 5 minutes; each strap anchor and docking tab 25 kg pull; each pouch strap and rail loop 15 kg for 1 minute, no stud release. Drop: packed to 7 kg, 5 drops from 1 m on each face, every pouch still attached. Zips: 2,000 cycles on the main zip. Press studs 1,000 cycles; docking buckles 500 cycles. Whistle audible at 1 m (100 dB). Tag colour-fast after 20 washes. Dimensions ±5 mm; empty weight 1.3 kg ±5%. Colour: all black parts match in daylight. Pre-shipment inspection by a third party to AQL 2.5 major / 4.0 minor.

Packing

Each bag: straps tucked, name card in its window, two spare cards in the lid sleeve, pouches flat inside and unfilled, in a recycled poly bag with a silica-gel sachet; no paper stuffing. 10 bags per 5-ply export carton (30 cartons for 300), carton under 20 kg, labelled BLACKOUT SOS · OPERATOR · BLUE · quantity · carton x of 30 · weight · dimensions · barcode.

Terms

Order: 300 Operators. Quote FOB Xiamen or Shenzhen and DDP London; HS 4202.92. Samples: swatch card and hardware set → pre-production sample → gold-seal sample signed by us and kept by the factory. Payment: 30% deposit, 70% after passed inspection. We supply vector artwork for the patch, tag, zip pull, labels and name card with the order to sample.